



TAVAONE EDUCATION · PINELLAS STEM RADIO LAB

BSA Radio Merit Badge

Session 2 — Digital, Devices & Badge Completion

Information · Hardware · Cell Phones · FT8 Contact · Blue Card Sign-Off (Reqs 4, 5, 7, 8-A(6), 9)

TavaOne Education · Pinellas STEM Radio Lab · Gulf Coast Contest Club



Session 2 Agenda

- Quick review of Session 1 material
- Req 4 — Information by Radio (modulation, digital standards, range)
- Req 5 — Radio Hardware and Services (block diagram, consumer devices, RFID)
- Req 7 — Cell Phones (cellular, Airplane Mode, GPS, emergencies, wireless charging)
- Req 8-A(6) — FT8 digital contact at the club station
- Req 9 — Career/hobby discussion with counselor
- Blue card review and counselor sign-off



Session 1 Quick Review — 60 Seconds

- Req 1: Three safety topics: grounding, electrical burns, Li-ion battery fires
- Req 2: EM spectrum from 300 kHz to 3 GHz — MF/HF/VHF/UHF/microwave
- Req 3: Three propagation modes — ground wave, sky wave, line-of-sight
- Req 6: FCC (US enforcement) vs. ITU (global rules); call signs; NATO phonetic alphabet
- Req 8-A (1–5): FCC amateur service; license classes; Q signals; emergency calls; HT/mobile/base/repeaters

- Any questions from Session 1 before we continue?

Requirement 4

Information by Radio — Modulation, Digital Standards & Data Rates



Req 4a — Five Ways Radio Carries Information

- CW (Morse Code): carrier keyed on/off → extremely narrow bandwidth → cuts through noise → still used for DX and weak-signal work
- AM (Amplitude Modulation): varies carrier strength → simple to receive → used for broadcast and aircraft voice → takes double the bandwidth of SSB
- FM (Frequency Modulation): varies carrier frequency → excellent audio quality → rejects interference well → used for VHF/UHF voice and FM broadcast
- SSB (Single Sideband): AM with carrier and one sideband removed → all power in useful signal → ~4× range of AM → standard HF voice mode
- Frequency Hopping: carrier jumps rapidly between many frequencies → impossible to intercept or jam → used in Bluetooth, military comms, and spread-spectrum Wi-Fi



Req 4b — Digital Radio Standards

Bluetooth & Wi-Fi

- Bluetooth: frequency-hopping at 2.4 GHz, hops 1,600×/second across 79 channels, short range (10–30 m), designed for low-power personal devices
- Wi-Fi 802.11: 2.4 GHz and 5 GHz, OFDM splits channel into hundreds of parallel subcarriers, 30–100 m indoor range, shared among all users
- Both are digital: information encoded as ones and zeros BEFORE transmission → allows error correction and encryption

5G and Why Digital Wins

- 5G: three tiers — sub-6 GHz for coverage, mmWave (24–100 GHz) for extreme speed in dense areas
- 5G uses massive MIMO + beamforming: aimed beams instead of broadcast
- Digital advantages over analog: error correction (FEC), encryption, more data per Hz, multiple channels multiplexed on one carrier (OFDM)
- Real-world 5G: average speeds far below peak; mmWave barely penetrates walls



Req 4c & 4d — Range vs. Encoding; Wi-Fi vs. Fiber

Range vs. Encoding Method

- More data rate = more bandwidth = needs stronger signal = shorter range
- CW/SSB/FT8: narrow bandwidth, most efficient → longest range per watt
- FT8 can decode signals 20 dB below the noise floor — contacts possible with 5 W across oceans
- High-speed Wi-Fi: wide bandwidth demands strong SNR → range shrinks rapidly with distance or walls

Wi-Fi vs. Wired/Fiber Data Rates

- Wi-Fi 6 peak: ~9.6 Gbps theoretical; real-world: 500 Mbps to 2 Gbps (close, clear)
- Fiber peak: 400+ terabits/sec on one strand using WDM; home service: 1–10 Gbps
- Why fiber wins: dedicated glass tube, no shared spectrum, no atmospheric noise, deterministic latency
- Wi-Fi latency varies under load; fiber latency is stable



Quick Check

SSB has roughly 4× the range of AM for the same power. Why?

A) SSB uses a higher frequency, which propagates better

B) SSB removes the carrier and one sideband, putting all power into the useful signal

C) SSB uses frequency hopping to avoid interference

D) SSB antennas are more efficient than AM antennas

Answer: B — SSB eliminates the carrier (which carries no information) and one of the two duplicate sidebands, concentrating all transmit power into a single usable sideband. Same information, less power wasted.

Requirement 5

Radio Hardware and Services



Req 5a — Block Diagram vs. Schematic Diagram

Block Diagram

- Shows major functional sections as labeled boxes
- Arrows indicate signal flow between sections
- No component-level detail (no resistors, capacitors)
- Used to: understand system organization, troubleshoot at system level, explain to non-engineers
- You can draw one without electronics knowledge

Schematic Diagram

- Shows every component using standardized electrical symbols
- Connections drawn as lines; you can trace every wire
- Used to: build circuits, calculate voltages/currents, repair equipment
- Requires training to read fluently
- A modern transceiver may have hundreds of schematic pages



Req 5b — Draw a Radio Station Block Diagram

- ACTIVITY: Draw your block diagram in your student pack now
- Microphone → Transmitter → [Power Amplifier] → Feedline → Antenna
- Antenna → Feedline → Receiver → Speaker
- (Transceiver = Transmitter + Receiver combined in one box)
- Key component functions:
 - Microphone: sound → electrical audio signal
 - Transmitter: audio → RF carrier with modulation
 - Amplifier (PA): increases RF power before antenna
 - Feedline: carries RF between radio and antenna (coax)
 - Antenna: electrical energy ↔ electromagnetic waves
 - Receiver: captures incoming RF → audio signal
 - Speaker: electrical audio → sound waves



Req 5c, 5d, 5e — Consumer Devices, Weather Radio & RFID

- Consumer devices using radio (5c): cell phone (4G/5G), car radio (AM/FM/satellite), Wi-Fi router (2.4/5 GHz), wireless earbuds (Bluetooth), GPS navigator (1.575 GHz), smart home devices (Zigbee/Wi-Fi), TV remote (IR or RF), contactless credit card (NFC at 13.56 MHz)
- NOAA Weather Radio (5d): 162.400–162.550 MHz, 7 channels, continuous NWS broadcasts, SAME coding = your county only alerts YOUR radio, covers 97% of US population, no license needed to receive
- RFID (5e): reader emits RF field → passive tag harvests energy from field → transmits its unique ID back. Active tags have their own battery. Uses: store anti-theft, library books, building access badges, pet microchips, passport chips, toll collection (E-ZPass/SunPass), supply chain tracking, tap-to-pay



Quick Check

A weather radio with SAME can be programmed to alert only for ____

- A) Any severe weather in the entire state
- B) Only the weather station's primary broadcast area
- C) Your specific county or FIPS region
- D) Any National Weather Service office within 200 miles

Answer: C — SAME (Specific Area Message Encoding) lets you program the radio to respond only to alerts for your FIPS county code. This prevents middle-of-the-night alerts for counties far away from you.

Requirement 7

Cell Phones — Cellular, Airplane Mode, GPS, Emergencies & Wireless Charging



Req 7a — Cellular vs. Broadcast vs. Amateur Radio

Cellular

- Two-way, digital, private
- User needs NO license (carrier is licensed)
- Requires towers, backhaul, billing — fails without infrastructure
- Coverage depends on tower proximity
- Monthly subscription cost

Broadcast vs. Amateur

- Broadcast: one-way only, licensed transmitters, free to receive, entertainment/news
- Amateur: two-way, requires FCC license to transmit, non-commercial, self-funded
- Amateur can operate with NO infrastructure — simplex, global HF
- Amateur: experimentation, emergency comms, worldwide contacts



Req 7b & 7c — Airplane Mode and Time/Location

Airplane Mode — What It Does and Why

- Disables ALL wireless transmitters: cellular, Wi-Fi, Bluetooth, NFC
- Can selectively re-enable Wi-Fi or BT after enabling Airplane Mode
- Why: avionics use sensitive RF receivers — cell phone TX bursts could cause interference, especially on older aircraft
- Cell phone in weak signal → increases TX power → more RF emissions
- FAA allows Wi-Fi at cruise altitude; cellular calling still restricted on US flights
- Also useful: save battery, reduce distractions, avoid roaming charges abroad

Time and Location via GPS

- Time sync: phone receives precise time from cellular network (NTP → GPS atomic clocks) — accurate to microseconds
- GPS location: signals from 24+ satellites at 20,200 km altitude
- Each satellite broadcasts its position and exact time; phone measures travel time from each
- Need signals from 4+ satellites for 3D position (lat, lon, elevation)
- Elevation accuracy: typically ± 30 ft, less accurate than horizontal



Req 7d & 7e — Cell Phones in Emergencies & Wireless Charging

- **EMERGENCY BENEFITS:** widespread coverage in populated areas · call 911 with any phone, even without a SIM · texts get through when voice calls fail · iPhone 14+ and some Android: satellite SOS with no cell coverage · WEA (Wireless Emergency Alerts) push to all phones in a cell area
- **EMERGENCY LIMITATIONS:** towers need power → batteries last 8–24 hrs → towers overload in disasters → no coverage in rural/remote areas. This is why ARES/RACES amateur radio operators staff EOCs.
- **WIRELESS CHARGING (Req 7e):** Inductive coupling — the charging pad coil carries AC → creates changing magnetic field → phone coil captures energy → converts back to DC. Standard: Qi (Wireless Power Consortium). MagSafe adds magnets for precise coil alignment at higher rates. Efficiency: ~80–85% vs. ~95% for wired.



Quick Check

Why do cell phone towers often fail after a major hurricane?

- A) Hurricanes only damage broadcast towers, not cellular towers
- B) Cell towers are small and easy to knock over with wind
- C) Towers need grid power; when the grid fails, backup batteries last only 8–24 hours
- D) The FCC shuts down towers during hurricanes to prevent interference

Answer: C — Cell towers run on backup batteries or generators when grid power fails. Most batteries only last 8–24 hours. When fuel runs out or generators fail, the towers go dark — exactly when people need communications most.

Requirement 8-A(6)

On-Air Digital Contact — FT8



About FT8 — The Most Popular Digital Mode

- Created by Joe Taylor (K1JT) and Steve Franke (K9AN) in 2017
 - Each transmission is EXACTLY 15 seconds — synchronized to GPS/internet time
 - Each message is 77 bits encoded as a 79-symbol sequence using Gaussian FSK
 - Software (WSJT-X) decodes signals 20 dB below the noise floor — below what human ears can hear
 - 5 W of power can make contacts across oceans — impossible with voice at the same power level
 - Most popular HF digital mode; on 20m (14.074 MHz) there are hundreds of stations on at any given moment
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- The tradeoff: FT8 is automated and very brief — you can't have a conversation. It proves a contact was made, but it's not a conversation mode like voice or CW.



FT8 Contact Procedure at the Club Station

- 1. Open WSJT-X — confirm your callsign, grid square (EL87), and radio connection
- 2. Set to 20m FT8 frequency: 14.074 MHz USB
- 3. Watch the DECODE pane — you will see many stations decoded in each 15-second window
- 4. Find a station calling CQ — double-click their line to begin the exchange
- 5. WSJT-X sends your response automatically in the next TX slot
- 6. The exchange takes 4–6 transmissions (1–2 minutes) — watch each message appear
- 7. When you see "73" confirmed, the contact is complete — WSJT-X logs it automatically
- 8. Copy the log entry into your student pack contact log



FT8 Message Sequence — What Each Line Means

- CQ W4GGJ EL87 → "Anyone listening? I'm W4GGJ in grid EL87."
 - W4GGJ KD9XYZ EM72 → "W4GGJ, I hear you. I'm KD9XYZ in grid EM72."
 - KD9XYZ W4GGJ -12 → "KD9XYZ, you're -12 dB. Here's my signal report."
 - W4GGJ KD9XYZ R-08 → "Roger — your report is -8 dB. Contact confirmed."
 - KD9XYZ W4GGJ RR73 → "Roger-roger, 73 — contact complete, goodbye."
 - W4GGJ KD9XYZ 73 → "73 — confirmed, done."
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- The dB numbers are the SNR (signal-to-noise ratio): -12 means 12 dB below the noise floor. Perfectly normal for FT8. Voice would be impossible at this level.

Requirement 9

Career / Hobby Report-Out — Counselor Discussion



Req 9 — Your Discussion with the Counselor

- This is a CONVERSATION — not a paper, not a memorized report
- Option 9a — Career Research:
 - Which career did you choose? What does the job involve day to day?
 - What training and education are required? What are the costs?
 - What are job prospects and salary ranges? Did this interest you?
- Option 9b — Ham Radio Hobby / Volunteer Path:
 - What hobby or volunteer role did you research? (ARES? RACES? POTA? Contesting?)
 - What license and training would you need?
 - What are startup costs? What organizations support this?
 - What are your short-term and long-term goals if you pursued it?

Blue Card Sign-Off

Final Review — All Nine Requirements



Blue Card Checklist — Every Requirement

- ✓ Req 1: Grounding (DC/AC/antenna/lightning), electrical burns, Li-ion battery fires
- ✓ Req 2a: EM spectrum chart drawn; 5 bands labeled; 8 services located
- ✓ Req 2b: Definition of radio; broadcast vs. 2-way; commercial vs. hobby discussed
- ✓ Req 3a: Propagation sketch (ground wave, sky wave, line-of-sight)
- ✓ Req 3b: WWV/WWVH band condition use explained
- ✓ Req 3c: DX vs. local explained
- ✓ Req 4: CW/AM/FM/SSB/freq-hopping; digital standards; range; Wi-Fi vs. fiber
- ✓ Req 5a–e: Block vs. schematic; block diagram drawn; consumer devices; weather radio; RFID
- ✓ Req 6: FCC vs. ITU; call signs; phonetic alphabet
- ✓ Req 7a–e: Cellular; Airplane Mode; GPS time/location; emergencies; wireless charging
- ✓ Req 8-A(1–5): FCC amateur service; licenses & VEs; Q signals; emergency calls; equipment
- ✓ Req 8-A(6): On-air contact logged with RST report
- ✓ Req 9: Career or hobby research discussed with counselor



Badge Complete!

Congratulations — you earned your Radio Merit Badge

What's Next?

- Interested in getting licensed? TavaOne Education runs Technician license courses on-site
- VE exam sessions held at Pinellas STEM Radio Lab — pass to get your own callsign
- Join the Gulf Coast Contest Club — club station access, mentorship, on-air events
- ARES / RACES — serve your community as an emergency communicator
- POTA (Parks on the Air) — take radio into the field and make contacts from national parks